When Fine-Tuning Fails: Lessons from MS MARCO Passage Ranking

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Problem Statement

- **Key Question** : Why does fine-tuning degrade performance on MS MARCO **[7]** passage ranking task?
- **Counterintuitive finding** : All fine-tuning approaches underperformed the base model (sentence-transformers/all-MiniLM-L6-v2 **[8]**)
- **Challenge**: Base model is a model pre-trained on 1 billion sentences including 9.1 million MS Marco examples. This makes it a highly optimised model for generating sentence embeddings.

Literature Review

- **Dual-Encoder architecture for passage ranking**

  Separate encoders are used for queries and passages with efficient similarity computation **[1]**. They are efficient as we can calculate and store the embeddings which makes inference fast.

- **Cross-Encoder architecture for passage ranking**

  Joint processing of query-passage pairs for nuanced relevance modeling is done **[2]**. This model is less time efficient as it processes each query and document pair at once.

Literature Review

- **LoRA (Low-Rank Adaptation)**

  Reduces trainable parameters while maintaining performance **[3]**

  Targets query/value projection matrices (~3.9% of total parameters)

- **Embedding Space Analysis**

  It is critical for retrieval performance

  UMAP: Preserves local and global structure for embedding analysis **[4]**

  Reveals degradation patterns beyond traditional metrics

  Essential for understanding fine-tuning effects on pre-optimized models

Literature Review

- **Research Gaps**

  Parameter-efficient methods need evaluation on saturated benchmarks

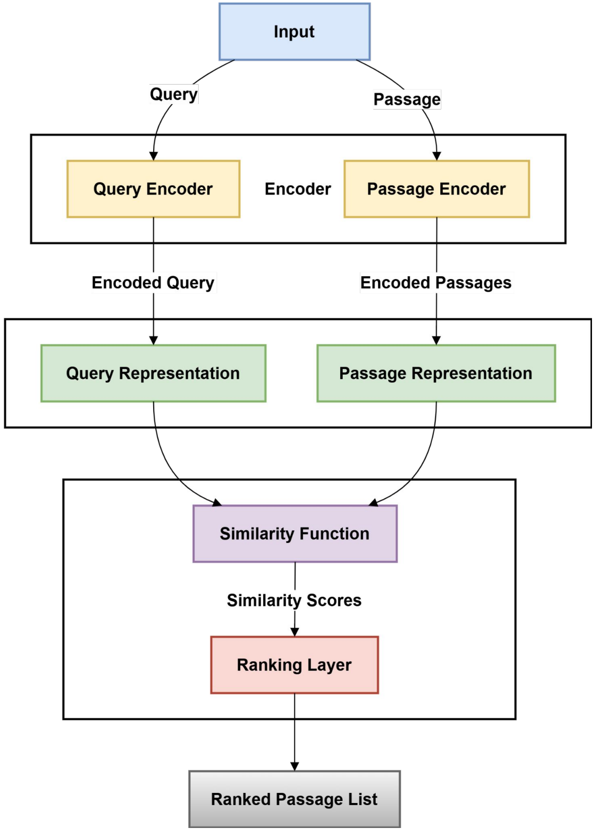
  Traditional fine-tuning assumptions may not hold for heavily pre-trained models

  Need for hybrid approaches combining statistical (BM25) and neural methods **[5]**

Research Context

- **Dataset**: MS MARCO - 8.8M passages, challenging retrieval benchmark
- **Base Model**: sentence-transformers/all-MiniLM-L6-v2 **[8]**
- **Extensive Pre-training**: 1 billion sentence pairs including 9.1M MS MARCO samples
- **Research Gap**: What happens when you fine-tune already optimized models?

Experimental Design



Experimental Design

- **5 model variants** :
  - Base SBERT (baseline)
  - Full Fine-tuning (Random negatives / Hard negatives)
  - LoRA Fine-tuning (Random negatives / Hard negatives)
- **Training Data**:

  1M random samples and 503k hard negatives vs base 1B samples in base model

  For building the hard negatives dataset we used our base model to retrieve top 200 passages for each training query. Randomly sampled a passage from range 51 to 100, excluding known positives to get our hard negative. This hard negative is semantically similar but not relevant.

- **Hardware**: NVIDIA A100 80GB GPU

Experimental Design

- **Loss function**:

  Triplet loss was used as the loss function, it is defined as :

  $L_{\textbf{triplet}} = \max(0, \mathrm{sim}(q, p^{-}) - \mathrm{sim}(q, p^{+}) + \alpha)$

  Where sim(q, p) is cosine similarity between queries and passage embeddings and α is the margin parameter. **[6]**

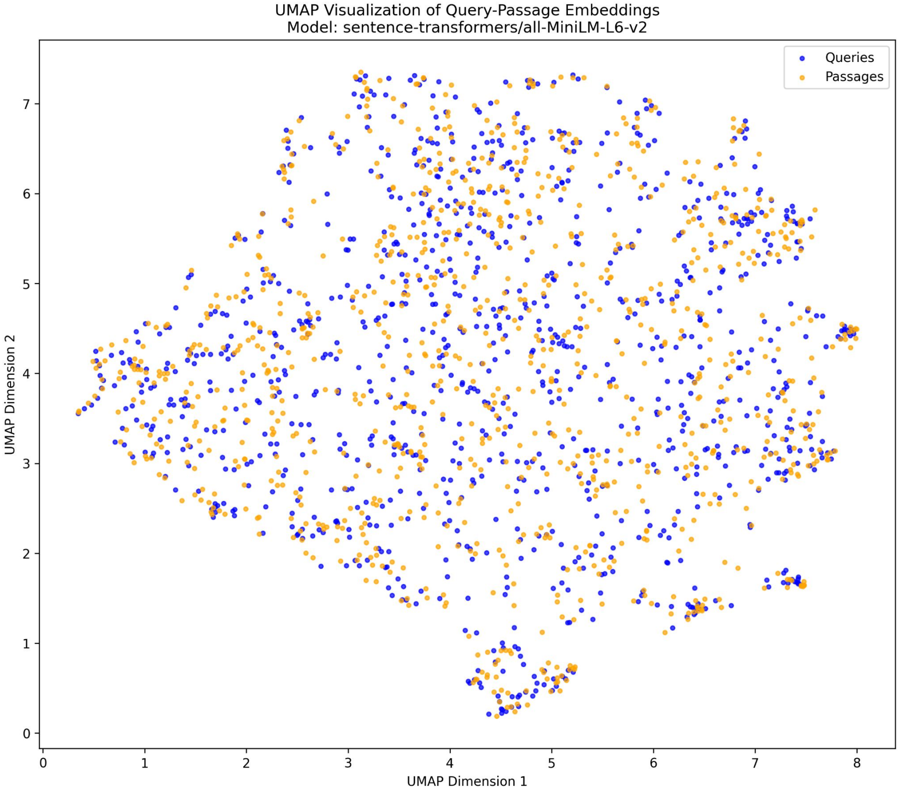
Key Results - Performance Degradation

MODEL	MRR@10	MRR@100	% change in MRR@10
Base SBERT	0.3026	0.3144	—
Full FT (Random)	0.2619	0.2723	-13.5%
Full FT (Hard)	0.2536	0.2632	-16.2%
LoRA FT (Random)	0.2557	0.2664	-15.5%
LoRA FT (Hard)	0.2050	0.2149	-32.3%

Key Results - Performance Degradation

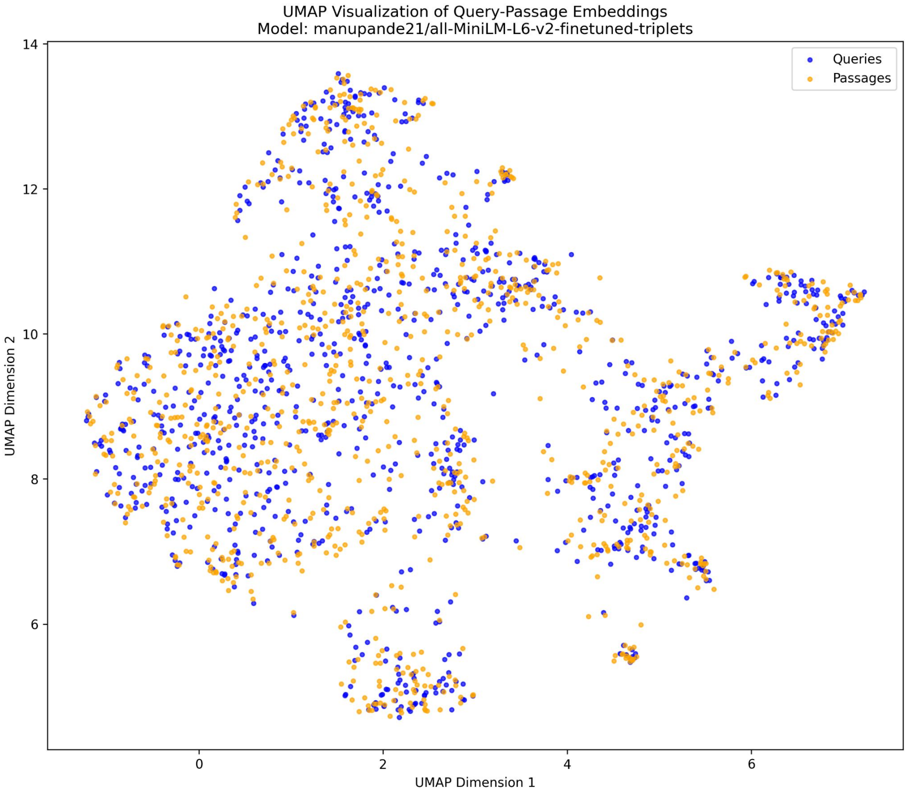
- **Worst Performer**: LoRA FT (Hard) with -32.3% degradation
- **Best Fine-tuned**: Full FT (Random), still -13.5% worse than base model
- **Universal Pattern**: No fine-tuning approach succeeded to overcome the benchmark set by the base model.

Embedding Space Analysis



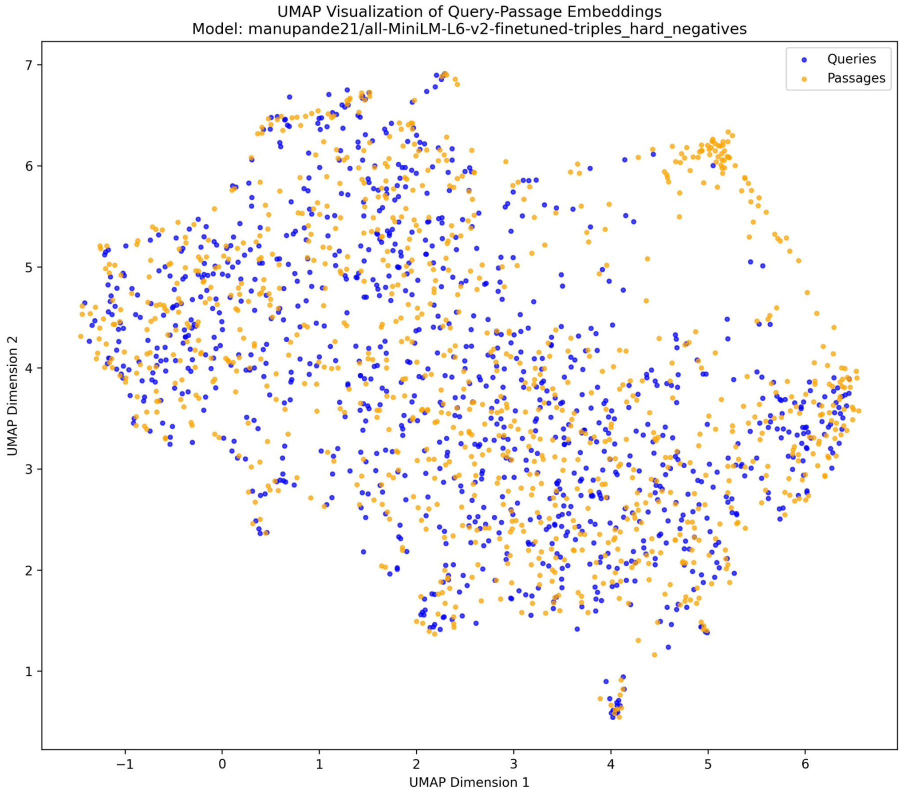
Base SBERT: Well-defined semantic clustering with natural boundaries and balanced distribution

Embedding Space Analysis



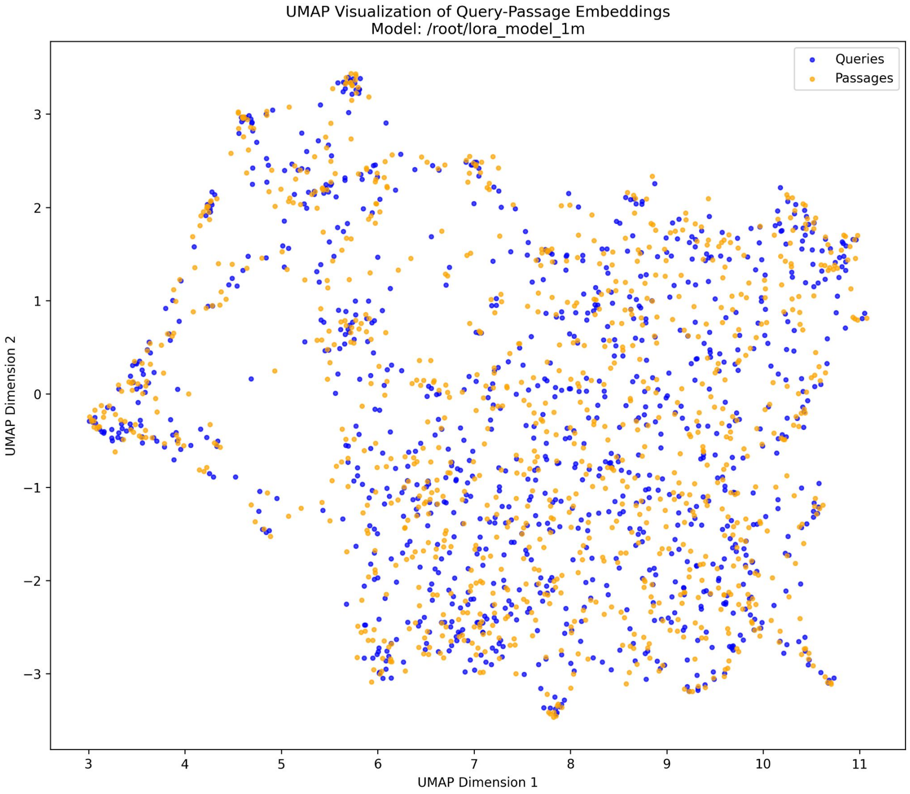
Full FT (Random): Distinct island formations
with preserved but altered clustering structure

Embedding Space Analysis



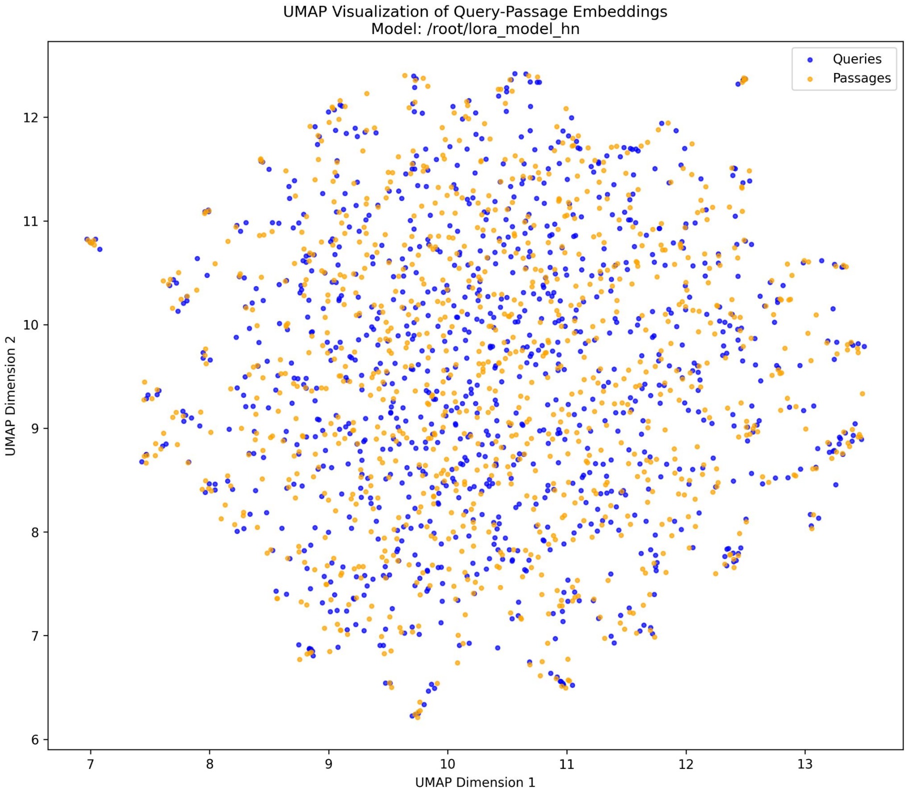
Full FT (Hard): Increased uniformity showing moderate embedding space flattening

Embedding Space Analysis



LoRA FT (Random):
Reduced semantic differentiation
with compressed clustering

Embedding Space Analysis



LoRA FT (Hard): Maximum uniformity demonstrating catastrophic embedding space collapse

Embedding Space Analysis

- **Key Insight**: Progressive degradation from structured clustering to uniformity
- **Visual Correlation**: Embedding space quality seems to directly correlate with the observed performance degradation
- **Mechanism**: Fine-tuning disrupts billion-scale learned representations thus adding more noise rather than information signal to the embedding space.

The Hard Negatives Paradox

- **Counterintuitive Finding**: Hard negatives perform worse than random negatives
- **Explanation**: Base model already optimized for semantic similarity
- **Implication**: Conventional negative sampling strategies fail on saturated models
- **LoRA Vulnerability**: Parameter efficient models show catastrophic sensitivity to hard negatives.

Computational Efficiency Surprise

MODEL	TIME (s)	% CHANGE FROM BASE	QUERIES PER SECOND
Base SBERT	303.92	—	32.9
Full FT (Random)	324.90	+6.9%	30.8
Full FT (Hard)	307.48	+1.2%	32.5
LoRA FT (Random)	574.92	+89.2%	17.4
LoRA FT (Hard)	598.69	+97.0%	16.7

Inference time analysis (10k queries)

Computational Efficiency Surprise

- **LoRA Performance**: 2× slower inference despite parameter efficiency
- **Hidden Costs**: Parameter efficiency ≠ computational efficiency
- **Deployment Impact**: Challenges assumptions about LoRA advantages

Key Insights & Implications

- **The Saturation Hypothesis**: Some benchmarks reach optimization limits
- **Scale Mismatch**: 1M fine-tuning samples vs 1B pre-training samples
- **Embedding Space Preservation**: Critical for maintaining performance
- **Research Paradigm Shift**: Need architectural innovation over parameter tuning

Future Directions

- **Beyond Parameter Tuning**: Focus on architectural innovations
- **Cross-encoder Models**: Joint query-document processing
- **Hybrid Systems**: Combine sparse (BM25) + dense retrieval
- **Alternative Loss Functions**: Beyond triplet loss, we can try out MultipleNegativesRankingLoss, Contrastive Loss **[6]**
- **Embedding Space Analysis**: Should become standard practice and should also include quantitative analysis

Takeaways

- **Main Finding**: Fine-tuning can harm heavily pre-trained models
- **Methodological Contribution**: Embedding space visualization as diagnostic tool
- **Practical Impact**: Challenges conventional transfer learning assumptions
- **Research Direction**: Architecture innovation > parameter optimization

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